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2027_IV_CSBS_DBMS Lab

PSGITECH_2027_DBMS_Week 9_COD

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Sarah is a data analyst working for a global company that provides products and services to customers worldwide. The company maintains a database with customer details, including their grades and assigned sales representatives. Sarah has been tasked with creating views that summarize customer data to help better understand customer distribution and performance.

Table Details:

Sample Input Records:

Table name: CUSTOMER

Task:

Create a view GRADECOUNT to return the distribution of customers by their grade. It should display the GRADE and the count of customers as GRADE_COUNT that belong to each grade.Create a view CUSTOMER_SUMMARY to provide a summary of the customer data, including the total number of customers as TOTAL_CUSTOMERS and the average grade as AVERAGE_GRADE across all customers.

Note: The user must write only the view queries to select the required records.

Answer

file.sql

```
CREATE VIEW GRADECOUNT AS
SELECT
   GRADE,
   COUNT(*) AS GRADE_COUNT
FROM
   CUSTOMER
GROUP BY
   GRADE;

CREATE VIEW CUSTOMER_SUMMARY AS
SELECT
   COUNT(*) AS TOTAL_CUSTOMERS,
   ROUND(AVG(GRADE), 4) AS AVERAGE_GRADE
FROM
   CUSTOMER;
```

Status : Correct ***Marks : 10/10***

2. Problem Statement

A sales team at a global corporation wants to analyze the performance of its salesmen across various cities. To assist with this analysis, the company needs to calculate the total commission earned by salesmen in each city and identify the salesmen who have a higher commission rate. The data is stored in the SALESMAN table.

Table Details:

Sample Input Records:

Table name: SALESMAN

Tasks to perform:

Create a view TOTAL_COMMISSION_PER_CITY to calculate the total commission earned by salesmen in each city. The view should display the CITY and the total commission as TOTAL_COMMISSION.Create a view SALES_MEN_WITH_HIGH_COMMISSION to display the salesmen who have a commission rate greater than 0.12. The result should include the SALESMAN_ID, NAME, CITY, and COMMISSION for the salesmen who meet this criterion.

Note: The user must write only the view queries to select the required records.

Answer

file.sql

```
CREATE VIEW TOTAL_COMMISSION_PER_CITY AS
SELECT
   CITY,
   ROUND(SUM(COMMISSION), 2) AS TOTAL_COMMISSION
FROM
   SALESMAN
GROUP BY
   CITY;

CREATE VIEW SALES_MEN_WITH_HIGH_COMMISSION AS
SELECT
   SALESMAN_ID,
   NAME,
   CITY,
```

```
   COMMISSION
FROM
   SALESMAN
WHERE
   COMMISSION > 0.12;
```

Status : Correct ***Marks : 10/10***

3. Problem Statement

Sophia is a business analyst working for an e-commerce company. She is tasked with analyzing the total purchase amounts across different orders. To do this, she needs to calculate the total amount for each order based on the quantity ordered and the price of each product. The data is stored in the ORDERDETAILS table.

Table Details:

Sample Input Records:

Table name: ORDERDETAILS

Task:

Write a query to create a view PURCHASE_AMOUNT_VIEW that shows the total amount spent for each order, grouping by ORDER_NUMBER. The view should display ORDER_NUMBER and TOTAL_AMOUNT.Write a query to create a view ORDER_SUMMARY_VIEW that shows the product count and total amount spent for each order, grouping by ORDER_NUMBER. The view should display ORDER_NUMBER, PRODUCT_COUNT, and TOTAL_AMOUNT.

Note: The user must write only the view queries to select the required records.

Answer

file.sql

```
CREATE VIEW PURCHASE_AMOUNT_VIEW AS
SELECT
```

```
    ORDER_NUMBER,
    ROUND(SUM(QUANTITY_ORDERED * PRICE_EACH), 2) AS TOTAL_AMOUNT
FROM
    ORDERDETAILS
GROUP BY
    ORDER_NUMBER;

CREATE VIEW ORDER_SUMMARY_VIEW AS
SELECT
    ORDER_NUMBER,
    COUNT(*) AS PRODUCT_COUNT,
    ROUND(SUM(QUANTITY_ORDERED * PRICE_EACH), 2) AS TOTAL_AMOUNT
FROM
    ORDERDETAILS
GROUP BY
    ORDER_NUMBER;
```

Status : Correct ***Marks : 10/10***

4. Problem Statement

Michael is a data analyst working for a global logistics company. The company maintains a database of stations across various cities and states, which are used to monitor the distribution of goods. Michael has been tasked with finding the city with the longest name to analyze the geographical distribution better.

Table Details:

Sample Input Records:

Table name: STATION

Task:

Create a view LONGESTCITY to return the name of the city with the longest name, along with the length of the city's name. The result should display the CITY and the LENGTH of the city's name, ordering the cities in descending order of their name length.Create a view SHORTESTCITY to return the name of the city with the shortest name, along with the length of the city's name. The result should display the CITY and the LENGTH of the city's name, ordering the cities in ascending order of their name length.

Note: The user must write only the view queries to select the required records.

Answer

file.sql

```
CREATE VIEW LONGESTCITY AS
SELECT
   CITY,
   LENGTH(CITY) AS LENGTH
FROM
   STATION
ORDER BY
   LENGTH DESC, CITY
LIMIT 1;

CREATE VIEW SHORTESTCITY AS
SELECT
   CITY,
   LENGTH(CITY) AS LENGTH
FROM
   STATION
ORDER BY
   LENGTH ASC, CITY
LIMIT 1;
```

Status : Correct ***Marks : 10/10***